

Ordinary Abelian Varieties over a Finite Field

P. Deligne

We give a down-to-earth description of the category of ordinary abelian varieties over a finite field $\mathbb{F}_q$. The result was inspired by Ihara [2, Chap. V]; see also [3].

1. Notation

Let p be a prime number, let $\mathbb{F}_p = \mathbb{Z}/(p)$, let $\overline{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$, and, for each power q of p , let $\mathbb{F}_q$ be the subfield of $\overline{\mathbb{F}}_p$ with q elements. For every algebraic extension k of $\mathbb{F}_p$, let $W_0(k)$ be the henselian discrete valuation ring, essentially of finite type over $\mathbb{Z}$, absolutely unramified, with residue field k . Let $W(k)$ be the ring of Witt vectors of k , i.e. the completion of $W_0(k)$; put $W = W(\overline{\mathbb{F}}_p)$, and fix an embedding $\varphi : W \hookrightarrow \mathbb{C}$.

Let $\mathbb{Z}(1)$ denote the subgroup $2\pi i\mathbb{Z}$ of $\mathbb{C}$. The exponential map gives an isomorphism

$$\mathbb{Z}(1) \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(1)(\mathbb{C}) = \varprojlim_m \mu_{\ell^m}(\mathbb{C}).$$

If A is an abelian variety, A^* denotes its dual. For every field k , $\bar{k}$ denotes an algebraic closure.

2. Ordinary abelian varieties

Let A be an abelian variety of dimension g over a field k of characteristic p . Recall that A is ordinary if the following equivalent conditions hold:

- (I) A has p^g points of order dividing p with values in $\bar{k}$;

(II) the Hasse–Witt matrix

$$F^* : H^1(A^{(p)}, \mathcal{O}_{A^{(p)}}) \longrightarrow H^1(A, \mathcal{O}_A)$$

is invertible;

(III) the identity component of the group scheme A_p , the kernel of multiplication by p , is of multiplicative type, hence geometrically isomorphic to a power of μ_p .

If $k = \mathbb{F}_q$, if F is the Frobenius endomorphism of A , and if $P_A(F; x)$ is its characteristic polynomial, these conditions are also equivalent to:

(IV) at least half of the roots of $P_A(F; x)$ in $\mathbb{Q}_p$ are p -adic units; equivalently, if $n = \dim A$, then the reduction of $P_A(F; x)$ modulo p is not divisible by x^{n+1} .

3. The canonical lift

Let A be an ordinary abelian variety over $\overline{\mathbb{F}}_p$. We denote by $\tilde{A}$ the Serre–Tate canonical lift of A to W . It depends functorially on A and is characterized by the fact that the p -divisible group $T_p(\tilde{A})$ over W attached to $\tilde{A}$ is the product of the p -divisible groups lifting, respectively, the identity component and the largest étale quotient of $T_p(A)$.

The canonical lift is also the unique lift of A to which every endomorphism of A lifts. Let $T(A)$ be the integral homology of the complex abelian variety $\tilde{A}_{\mathbb{C}}$ obtained from $\tilde{A}$ by extending scalars from W to $\mathbb{C}$ through φ :

$$T(A) = H_1(\tilde{A}_{\mathbb{C}}, \mathbb{Z}).$$

The canonical lift descends uniquely to $W_0(\overline{\mathbb{F}}_p)$, so $\tilde{A}_{\mathbb{C}}$ depends only on A and on the restriction of φ to $W_0(\overline{\mathbb{F}}_p)$. The free $\mathbb{Z}$ -module $T(A)$ has rank $2 \dim A$ and is functorial in A . If $\ell \neq p$, there is a functorial identification

$$T(A) \otimes \mathbb{Z}_{\ell} = T_{\ell}(A). \tag{1}$$

The canonical lift of the dual abelian variety A^* is the dual of $\tilde{A}$. Thus $(\tilde{A}_{\mathbb{C}})^* = \tilde{A}^*_{\mathbb{C}}$, and $T(A)$ and $T(A^*)$ are in perfect duality with values in $\mathbb{Z}(1)$:

$$T(A) \otimes T(A^*) \longrightarrow \mathbb{Z}(1). \quad (2)$$

The use of $\mathbb{Z}(1)$, rather than $\mathbb{Z}$, is essential for a theory invariant under complex conjugation. The pairings (2) are compatible, via (1), with the pairings

$$T_{\ell}(A) \otimes T_{\ell}(A^*) \longrightarrow \mathbb{Z}_{\ell}(1).$$

A morphism $\xi : A \rightarrow A^*$ defines a polarization of A if and only if $\xi_{\mathbb{C}} : \tilde{A}_{\mathbb{C}} \rightarrow \tilde{A}^*_{\mathbb{C}}$ defines a polarization of $\tilde{A}_{\mathbb{C}}$.

Put

$$T'_p(A) = \text{Hom}(\mathbb{Q}_p/\mathbb{Z}_p, A(\overline{\mathbb{F}}_p)), \quad T''_p(A) = \text{Hom}_{\mathbb{Z}_p}(T'_p(A^*), \mathbb{Z}(1) \otimes \mathbb{Z}_p).$$

These $\mathbb{Z}_p$ -modules are covariant functors of A . By definition of the canonical lift, the p -divisible group $T_p(\tilde{A})$ is the sum of the constant pro-étale group $T'_p(A)$ and the Cartier dual of $T'_p(A^*)$. For every morphism $u : A \rightarrow B$, the induced morphism $T_p(\tilde{A}) \rightarrow T_p(\tilde{B})$ identifies with the sum of $u|_{T'_p(A)} : T'_p(A) \rightarrow T'_p(B)$ and the Cartier transpose of ${}^t u|_{T'_p(B^*)} : T'_p(B^*) \rightarrow T'_p(A^*)$. Over $\mathbb{C}$ there is a canonical isomorphism $\mathbb{Z}(1)/p^n \mathbb{Z}(1) \simeq \mu_{p^n}$, hence a functorial decomposition

$$T_{(p)}(A) := T(A) \otimes \mathbb{Z}_p = T'_p(A) \oplus T''_p(A). \quad (3)$$

4. Finite subgroups

If $\psi : X \rightarrow Y$ is an isogeny of complex abelian varieties, the exact homotopy sequence reduces to a short exact sequence

$$0 \longrightarrow H_1(X) \longrightarrow H_1(Y) \longrightarrow \text{Ker}(\psi) \longrightarrow 0.$$

Thus quotient abelian varieties of X by finite subgroups, and the finite subgroups themselves, are in bijection with the overlattices of $H_1(X) \otimes \mathbb{Q}$ containing $H_1(X)$.

Let A be ordinary over $\overline{\mathbb{F}}_p$. If n is prime to p , the finite group subschemes of order n of A , of $\tilde{A}$, and of $\tilde{A}_{\mathbb{C}}$ correspond bijectively; they also correspond to overlattices R of $T(A)$ such that $[R : T(A)] = n$.

Put

$$V'_p(A) = T'_p(A) \otimes \mathbb{Q}_p, \quad V''_p(A) = T''_p(A) \otimes \mathbb{Q}_p.$$

Finite group subschemes of A of order p^k are products of an étale subgroup and an infinitesimal subgroup. Étale subgroups of order p^k correspond to those subgroups of order p^k of $\tilde{A}_{\mathbb{C}}$ for which the corresponding overlattice R of $T(A)$ is contained in $T_{(p)}(A) + V'_p(A)$. By duality, infinitesimal subgroups correspond to overlattices R of $T(A)$ which are p -isogenous to $T(A)$, i.e. for which $[R : T(A)]$ is a power of p , and are contained in $T_{(p)}(A) + V''_p(A)$.

Thus finite subgroups of A , or quotient abelian varieties of A , correspond bijectively to overlattices R of $T(A)$ satisfying

$$R \otimes \mathbb{Z}_p = (R \otimes \mathbb{Z}_p \cap V'_p) + (R \otimes \mathbb{Z}_p \cap V''_p). \quad (4)$$

5. The quotient by the infinitesimal part

In particular, $A^{(p)}$, the quotient of ordinary A by its largest infinitesimal subgroup killed by p , is defined by the overlattice $T(A)^{(p)}$ of $T(A)$, p -isogenous to $T(A)$, such that

$$T(A)^{(p)} \otimes \mathbb{Z}_p = T'_p(A) + \frac{1}{p}T''_p(A).$$

6. Frobenius

Let A be an abelian variety over $\mathbb{F}_q$, and let $F : x \mapsto x^q$ be its Frobenius endomorphism. Recall that A is uniquely determined

by the pair $(\overline{A}, F)$ obtained from (A, F) after extending scalars from $\mathbb{F}_q$ to $\overline{\mathbb{F}}_q$. The endomorphism F of $\overline{A}$ factors through the relative Frobenius $Fr^{(q)} : \overline{A} \rightarrow \overline{A}^{(q)}$ and an isomorphism $F' : \overline{A}^{(q)} \rightarrow \overline{A}$.

If A is ordinary, we write $T(A)$ for the $\mathbb{Z}$ -module $T(\overline{A})$ endowed with the endomorphism F induced by Frobenius. By the preceding discussion and (3), the lattices $T(A)$ and $F(T(A))$ are p -isogenous and

$$F(T'_p(A)) = T'_p(A), \quad (5)$$

$$F(T''_p(A)) = qT''_p(A). \quad (6)$$

7. The theorem

Theorem 1. *The functor*

$$A \longmapsto (T(A), F)$$

is an equivalence between the category of ordinary abelian varieties over $\mathbb{F}_q$ and the category of finite-rank free $\mathbb{Z}$ -modules T endowed with an endomorphism F satisfying:

- (a) *F is semisimple, and its eigenvalues have complex absolute value $q^{1/2}$;*
- (b) *at least half of the roots in $\mathbb{Q}_p$ of the characteristic polynomial of F are p -adic units; equivalently, if T has rank d , the reduction of $P_T(F; x)$ modulo p is not divisible by $x^{[d/2]+1}$;*
- (c) *there exists an endomorphism V of T such that $FV = q$.*

If (a) holds, then (b) and (c) are equivalent to:

- (d) *$T \otimes \mathbb{Z}_p$ has an F -stable decomposition into two $\mathbb{Z}_p$ -submodules T'_p and T''_p of the same dimension, such that $F|T'_p$ is invertible and $F|T''_p$ is divisible by q .*

Proof. (A) We first show that (a), (b), and (c) imply (d). If α is a complex eigenvalue of F , then q/α is another eigenvalue with the same multiplicity, and $\alpha(q/\alpha) = q$. Apart from possible values $\pm q^{1/2}$, the eigenvalues of F in $\mathbb{C}$, and therefore in $\overline{\mathbb{Q}_p}$, fall into pairs α and q/α . The two roots in such a pair cannot both be p -adic units. By (b), $\pm q^{1/2}$ is not an eigenvalue; half of the eigenvalues in $\mathbb{Q}_p$ are units, say $\alpha_1, \dots, \alpha_{d/2}$, and the other half are $\beta_i = q/\alpha_i$. Let $T_{(p)} = T \otimes \mathbb{Z}_p$, $V_p = T \otimes \mathbb{Q}_p$, let V'_p be the kernel of $\prod(F - \alpha_i)$, and let V''_p be the kernel of $\prod(F - \beta_i)$. Then $V_p = V'_p \oplus V''_p$. If T'_p is the projection of $T_{(p)}$ onto V'_p and $T''_p = T_{(p)} \cap V''_p$, then $T_{(p)} = T'_p \oplus T''_p$, giving (d).

(B) Full faithfulness. For abelian varieties A and B over $\mathbb{F}_q$, let

$$\psi : \text{Hom}(A, B) \longrightarrow \text{Hom}_F(T(A), T(B))$$

be the natural map. Tate's theorem, together with (1), shows that

$$\psi_\ell : \text{Hom}(A, B) \otimes \mathbb{Z}_\ell \longrightarrow \text{Hom}_F(T(A), T(B)) \otimes \mathbb{Z}_\ell$$

is an isomorphism for $(\ell, p) = 1$. Hence $\psi \otimes \mathbb{Q}$ is an isomorphism. Since $\text{Hom}(A, B)$ is torsion-free, ψ is injective. If $u : A \rightarrow B$ is such that $T(u)$ is divisible by n , then $u_{\mathbb{C}} : \tilde{A}_{\mathbb{C}} \rightarrow \tilde{B}_{\mathbb{C}}$ is divisible by n , and so is the lift $\tilde{u} : \tilde{A} \rightarrow \tilde{B}$ at the generic point of W . The kernel of multiplication by n is flat over W , so $\tilde{u}$ vanishes on that kernel; therefore $\tilde{u}$ and u are divisible by n . Thus ψ is bijective.

(C) Necessity. For $(T(A), F)$, condition (a) follows from Weil. Condition (d), and hence (b) and (c), follows from (5) and (6).

(D) Isogenies. Let (T_0, F) satisfy (a) and (d), and let T be an F -stable lattice in $T_0 \otimes \mathbb{Q}$ satisfying (d). Suppose (T_0, F) comes from an abelian variety A over $\mathbb{F}_q$. After multiplying T by a rational number, we may assume $T \supset T_0$. Condition (d) implies (4); T therefore defines a subgroup H of A , defined over $\mathbb{F}_q$, and $(T, F) = T(A/H)$.

(E) Essential surjectivity. Passing to isogeny classes, the

functor T sends ordinary abelian varieties over $\mathbb{F}_q$ to finite-dimensional $\mathbb{Q}$ -vector spaces with an automorphism F satisfying (a) and (b). By (D), it remains to prove essential surjectivity on simple objects. For every simple object (V, F) in the image category, Honda's theorem gives an abelian variety A over $\mathbb{F}_q$ whose Frobenius characteristic polynomial is a power of that of F . The criterion in Section 2 shows that A is ordinary. Moreover $T(A) \otimes \mathbb{Q}$ is a sum of copies of (V, F) ; by (B), the isogeny class of A is a sum of copies of an abelian variety B with $T(B) \otimes \mathbb{Q} = (V, F)$. $\square$

8. The complex structure

Let (T, F) satisfy the hypotheses of the theorem, let $2g = \text{rank } T$, and let A be the corresponding abelian variety over $\mathbb{F}_q$. For the complex abelian variety $\tilde{A}_{\mathbb{C}}$ obtained from the canonical lift,

$$T = H_1(\tilde{A}_{\mathbb{C}}, \mathbb{Z}).$$

Thus $T \otimes \mathbb{R}$ is identified with the Lie algebra of $\tilde{A}_{\mathbb{C}}$ and has a complex structure. According to J.-P. Serre, this complex structure can be reconstructed from T , F , and the restriction of φ as follows.

Proposition 1. *The complex structure on $T \otimes \mathbb{R}$ defined above is characterized by:*

- (I) *the endomorphism F is $\mathbb{C}$ -linear;*
- (II) *if v is the valuation of the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ extending the valuation of $W_0(\overline{\mathbb{F}}_p)$, then the valuations of the g eigenvalues of this endomorphism are strictly positive.*

Condition (I) is clear. Condition (II) follows from the fact that the action of F on the Lie algebra of $\tilde{A}$ is congruent to zero modulo p . Uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) for (T, F) .

References

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